Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Structure decision-making problems under uncertainty, calculate and organise payoffs, and apply the Maximax and Maximin criteria to compare alternatives and justify decisions.

Criteria assessment

This task sheet assesses criteria C1 through C4. Each problem assesses the criteria indicated in its title. The number of points assigned to each task is shown at the end of the item. Partial credit may be awarded when the correct procedure is shown but arithmetic errors are made. Answers that do not show the required procedure will not receive credit. A minimum of 4 points is required to pass each criterion.

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C2 Organises decision problems using payoff tables, placing alternatives, states of nature, and payoffs in a clear structure for comparison.

C3 Calculates profits by combining revenues with fixed and variable costs across different states of nature, determining the corresponding payoffs for each alternative, and organising the results in payoff tables for comparison.

C4 Applies the Maximin and Maximax criteria to make optimistic or conservative decisions without probabilities, selecting the alternative with the highest possible payoff or the best worst-case, interpreting risks or limitations.

1 Community Festival Beverage Booth [C1-1 C2-2]

A community festival company must decide which beverage booth to operate before knowing how strong customer attendance will be. The company can choose either a basic booth or a premium booth, while festival attendance may turn out to be high or low. If the company chooses the basic booth, it expects to earn a profit of sixteen thousand US dollars when attendance is high and ten thousand US dollars when attendance is low. If it chooses the premium booth, it expects to earn a profit of twenty-nine thousand US dollars when attendance is high and five thousand US dollars when attendance is low.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**
C2 Construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Basic booth	**A2:** Premium booth
Statesofnature	**S1:** High attendance	**S2:** Low attendance
Events	Actual festival attendance after the booth decision	
Consequences	Profit in thousand US dollars for each alternative–state pair	

C2 - Payoff table [2]

Each stated consequence is placed at the intersection of its alternative and state:

$$P(A_1, S_1) = 16, \quad P(A_1, S_2) = 10,$$
$$P(A_2, S_1) = 29, \quad P(A_2, S_2) = 5.$$

Thus, in thousand US dollars, the completed payoff table is

Alternative	High S_1	Low S_2
Basic booth A_1	16	10
Premium booth A_2	29	5



2 Local Delivery Fleet [C1-1 C2-3]

A local delivery firm must choose between a bicycle fleet and an electric-van fleet before traffic conditions are known. Traffic may be clear or congested. The bicycle fleet earns 4 thousand CNY per completed delivery batch and is expected to complete 14 delivery batches in clear traffic and 10 in congested traffic. The electric-van fleet earns 6 thousand CNY per completed delivery batch and is expected to complete 12 delivery batches in clear traffic and 8 in congested traffic.

Clear traffic also adds a fixed scheduling bonus of 5 thousand CNY to either fleet, while congestion creates a fixed loss of 4 thousand CNY.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C2 Construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Bicycle fleet	**A2:** Electric-van fleet
Statesofnature	**S1:** Clear traffic	**S2:** Congested traffic
Events	Actual traffic condition after the fleet decision	
Consequences	Profit in thousand CNY for each alternative–state pair	

C2 - Payoff table [3]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected completed delivery batches for each fleet under each traffic condition	$q(A_i, S_j)$	Bicycle–clear: 14 batches; Bicycle–congested: 10 batches; Electric van–clear: 12 batches; Electric van–congested: 8 batches
Variable, alternative dependent	Earnings per completed delivery batch from the chosen fleet	$v^A(A_i)$	Bicycle fleet: 4 thousand CNY/batch; Electric-van fleet: 6 thousand CNY/batch
Variable, state dependent	Additional contribution per completed delivery batch from the traffic condition	$v^S(S_j)$	Clear traffic: 0 thousand CNY/batch; Congested traffic: 0 thousand CNY/batch
Fixed, state dependent	Fixed scheduling bonus or loss associated with the traffic condition	$f^S(S_j)$	Clear traffic: 5 thousand CNY; Congested traffic: -4 thousand CNY

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)v^A(A_i) + f^S(S_j).$$

Calculations (thousand CNY).

Fleet	Clear traffic	Congested traffic
Bicycle fleet	$14(4) + 5 = 61$	$10(4) - 4 = 36$
Electric-van fleet	$12(6) + 5 = 77$	$8(6) - 4 = 44$

Final payoff table (profit in thousand CNY).

Fleet	Clear traffic	Congested traffic
Bicycle fleet	61	36
Electric-van fleet	77	44



3 Regional Food Delivery Service [C1-2 C2-5]

A regional food delivery company must choose between a standard delivery service and an express delivery service before customer demand is known. Demand may be high or low. The standard service earns 3 thousand GBP per completed delivery order and also provides a fixed operating contribution of 5 thousand GBP. The express service earns 5 thousand GBP per completed delivery order and provides a fixed operating contribution of 8 thousand GBP. High demand adds another 2 thousand GBP per completed order to either service, while low demand adds 1 thousand GBP per completed order. The standard service is expected to complete 15 orders under high demand and 9 orders under low demand; the express service is expected to complete 11 orders under high demand and 7 orders under low demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**
C2 Construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	**A1:** Standard delivery service	**A2:** Express delivery service
Statesofnature	**S1:** High customer demand	**S2:** Low customer demand
Events	Actual customer demand after the delivery service is selected	
Consequences	Profit in thousand GBP for each alternative–state pair	

C2 - Payoff table [5]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Expected completed delivery orders for each service under each demand level	$q(A_i, S_j)$	Standard–high: 15 orders; Standard–low: 9 orders; Express–high: 11 orders; Express–low: 7 orders
Variable, alternative dependent	Contribution per completed delivery order from the chosen service	$v^A(A_i)$	Standard service: 3 thousand GBP/order; Express service: 5 thousand GBP/order
Variable, state dependent	Additional contribution per completed order from customer demand	$v^S(S_j)$	High demand: 2 thousand GBP/order; Low demand: 1 thousand GBP/order
Fixed, alternative dependent	Fixed operating contribution from the chosen service	$f^A(A_i)$	Standard service: 5 thousand GBP; Express service: 8 thousand GBP

Payoff equation.

$$P_{ay}(A_i, S_j) = q(A_i, S_j)\left[v^A(A_i) + v^S(S_j)\right] + f^A(A_i).$$

Calculations (thousand GBP).

Service	High customer demand	Low customer demand
Standard service	$15(3+2)+5=80$	$9(3+1)+5=41$
Express service	$11(5+2)+8=85$	$7(5+1)+8=50$

Final payoff table (profit in thousand GBP).

Service	High customer demand	Low customer demand
Standard service	80	41
Express service	85	50



4 Local Caf'e Investment [C1-1 C3-2]

A small caf'e must decide whether to launch an artisan-dessert line or keep the current menu before foot traffic is known. Traffic may be high or low. All values are in million COP. Launching the dessert line produces revenue and cost of 58 and 16 under high traffic, and 27 and 19 under low traffic. Keeping the current menu produces revenue and cost of 49 and 21 under high traffic, and 41 and 21 under low traffic.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[2]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Launch dessert line	**A2:** Keep current menu
Statesofnature	**S1:** High traffic	**S2:** Low traffic
Events	Actual foot traffic after the menu decision	
Consequences	Profit in million COP for each alternative–state pair	

C3 - Calculating profit and building the payoff table [2]

Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Revenue, alternative–state dependent	Total revenue generated by each menu decision under each traffic condition	$R(A_i, S_j)$	Dessert–high: 58 million COP; Dessert–low: 27 million COP; Current menu–high: 49 million COP; Current menu–low: 41 million COP
Cost, alternative–state dependent	Total cost associated with each menu decision under each traffic condition	$C(A_i, S_j)$	Dessert–high: 16 million COP; Dessert–low: 19 million COP; Current menu–high: 21 million COP; Current menu–low: 21 million COP

Profit equation.

$$P_{ay}(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Calculations (million COP).

Alternative and state	Revenue	Cost	Profit
Dessert line–high traffic	58	16	$58 - 16 = 42$
Dessert line–low traffic	27	19	$27 - 19 = 8$
Current menu–high traffic	49	21	$49 - 21 = 28$
Current menu–low traffic	41	21	$41 - 21 = 20$

Final payoff table (profit in million COP).

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep current menu	28	20



5 Custom Furniture Product Line [C1-1 C3-3]

A custom-furniture workshop must choose between producing dining tables and producing office desks before market conditions are known. Conditions may be strong demand with favourable material prices or weak demand with higher material prices. The respective quantities are 7 and 4 production lots for dining tables, and 6 and 3 lots for office desks. Revenue per lot depends on the product: 14 thousand EUR for dining tables and 16 thousand EUR for office desks. Cost per lot depends on the state: 6 thousand EUR under strong demand and favourable prices, and 9 thousand EUR under weak demand and higher prices.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[1]**

C3 Use revenue and costs to construct the payoff table. **[3]**

C1 - Characterization [1]

Element	Description	
Alternatives	**A1:** Produce dining tables	**A2:** Produce office desks
Statesofnature	**S1:** Strong demand and favourable material prices	**S2:** Weak demand and higher material prices
Events	Market conditions revealed after the production decision	
Consequences	Profit in thousand EUR for each product–market-state pair	

C3 - Profit and payoff table [3]



Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Production lots for each product under each market condition	$q(A_i, S_j)$	Tables–strong: 7; Tables–weak: 4; Desks–strong: 6; Desks–weak: 3
Variable, alternative dependent	Revenue per production lot	$r(A_i)$	Dining tables: 14 thousand EUR/lot; Office desks: 16 thousand EUR/lot
Variable, state dependent	Cost per production lot	$c(S_j)$	Strong demand: 6 thousand EUR/lot; Weak demand: 9 thousand EUR/lot

Profit equation.

$$R = q(A_i, S_j)r(A_i), \qquad C = q(A_i, S_j)c(S_j), \qquad P_{ay}(A_i, S_j) = R - C.$$

Calculations (thousand EUR).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$7(14) = 98$	$7(6) = 42$	$98 - 42 = 56$
(A_1, S_2)	$4(14) = 56$	$4(9) = 36$	$56 - 36 = 20$
(A_2, S_1)	$6(16) = 96$	$6(6) = 36$	$96 - 36 = 60$
(A_2, S_2)	$3(16) = 48$	$3(9) = 27$	$48 - 27 = 21$

Final payoff table (profit in thousand EUR).

Alternative	Strong demand, favourable prices	Weak demand, higher prices
Produce dining tables	56	20
Produce office desks	60	21

6 Farm Subscription Plans [C1-2 C3-5]

A farm cooperative must choose between a standard produce box and a premium produce box before seasonal demand is known. Demand may be strong or weak. Quantities, in hundreds of subscriptions, are 8 and 5 for the standard box and 6 and 4 for the premium box under strong and weak demand, respectively. The plan-based revenue rates are 9 and 12 thousand USD per hundred subscriptions for the standard and premium boxes. Strong demand adds a state-based revenue rate of 3 thousand USD per hundred subscriptions; weak demand adds 1. The plan-based cost rates are 4 and 7 thousand USD per hundred subscriptions, respectively. The fixed distribution costs are 10 thousand USD under strong demand and 14 thousand USD under weak demand.

Questions/Tasks.

C1 Characterize the key elements of the decision-making situation. **[2]**
C3 Use revenue and costs to construct the payoff table. **[5]**

C1 - Characterization [2]

Element	Description	
Alternatives	**A1:** Offer a standard produce box	**A2:** Offer a premium produce box
Statesofnature	**S1:** Strong seasonal demand	**S2:** Weak seasonal demand
Events	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Profit in thousand USD for each plan–market-state pair	

C3 - Profit and payoff table [5]



Relevant values.

Abstract component	Context component	Math symbol	Corresponding value
Quantity, alternative–state dependent	Subscription lots for each plan under each demand level	$q(A_i, S_j)$	Standard–strong: 8; Standard–weak: 5; Premium–strong: 6; Premium–weak: 4 hundreds of subscriptions
Variable, alternative dependent	Plan-based revenue rate per hundred subscriptions	$r^A(A_i)$	Standard box: 9 thousand USD/100 subscriptions; Premium box: 12 thousand USD/100 subscriptions
Variable, state dependent	Additional revenue rate per hundred subscriptions from seasonal demand	$r^S(S_j)$	Strong demand: 3 thousand USD/100 subscriptions; Weak demand: 1 thousand USD/100 subscriptions
Variable, alternative dependent	Plan-based cost rate per hundred subscriptions	$c^A(A_i)$	Standard box: 4 thousand USD/100 subscriptions; Premium box: 7 thousand USD/100 subscriptions
Fixed, state dependent	Fixed distribution cost under each demand condition	$f_c(S_j)$	Strong demand: 10 thousand USD; Weak demand: 14 thousand USD

Profit equation.

$$R = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right], \qquad C = q(A_i, S_j)c^A(A_i) + f_c(S_j), \qquad P_{ay}(A_i, S_j) = R - C.$$

Calculations (thousand USD).

Cell	Revenue	Cost	Profit
(A_1, S_1)	$8(9+3) = 96$	$8(4) + 10 = 42$	$96 - 42 = 54$
(A_1, S_2)	$5(9+1) = 50$	$5(4) + 14 = 34$	$50 - 34 = 16$
(A_2, S_1)	$6(12+3) = 90$	$6(7) + 10 = 52$	$90 - 52 = 38$
(A_2, S_2)	$4(12+1) = 52$	$4(7) + 14 = 42$	$52 - 42 = 10$

Final payoff table (profit in thousand USD).

Alternative	Strong seasonal demand	Weak seasonal demand
Standard produce box	54	16
Premium produce box	38	10

7 New Device Launch Strategy [C4-2]

A technology company must choose a strategy for launching a new wearable device. Market demand could be high, medium, or low. All payoffs are in thousands of dollars.

Alternative	Demand		
	High	Medium	Low
Premium launch	90	38	-25
Standard launch	65	45	20
Economy launch	42	34	27
Delay launch	18	18	18

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_3$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	90	65	42	18	$A^*_{\text{Maximax}} = A_1$
Min	-25	20	27	18	$A^*_{\text{Maximin}} = A_3$

Maximax

$$\begin{aligned} M_1 &= \max\{90, 38, -25\} = 90 \\ M_2 &= \max\{65, 45, 20\} = 65 \\ M_3 &= \max\{42, 34, 27\} = 42 \\ M_4 &= \max\{18, 18, 18\} = 18 \end{aligned}$$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Premium launch)}$$

Maximin

$$\begin{aligned} m_1 &= \min\{90, 38, -25\} = -25 \\ m_2 &= \min\{65, 45, 20\} = 20 \\ m_3 &= \min\{42, 34, 27\} = 27 \\ m_4 &= \min\{18, 18, 18\} = 18 \end{aligned}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_3 \quad \text{(Economy launch)}.$$

8 Restaurant Expansion Strategy **[C4-2]**

A restaurant chain must choose a strategy for expanding its operations. Customer demand could be high, medium, or low. All payoffs are in thousands of dollars.

	Customer demand		
Alternative	High	Medium	Low
Large expansion	85	32	-18
Moderate expansion	62	44	16
Small expansion	45	36	24
No expansion	20	20	20

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**

Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_3$

Maximax

$$\begin{aligned} M_1 &= \max\{85, 32, -18\} = 85 \\ M_2 &= \max\{62, 44, 16\} = 62 \\ M_3 &= \max\{45, 36, 24\} = 45 \\ M_4 &= \max\{20, 20, 20\} = 20 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{85, 32, -18\} = -18 \\ m_2 &= \min\{62, 44, 16\} = 16 \\ m_3 &= \min\{45, 36, 24\} = 24 \\ m_4 &= \min\{20, 20, 20\} = 20 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	85	62	45	20	$A^*_{\text{Maximax}} = A_1$
Min	-18	16	24	20	$A^*_{\text{Maximin}} = A_3$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Large expansion)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_3 \quad \text{(Small expansion).}$$

9 Renewable Energy Investment Strategy **[C4-2]**

A renewable-energy company must choose a strategy for developing a new solar project. Market conditions could be very favourable, favourable, stable, or unfavourable. All payoffs are in thousands of dollars.

	Market conditions			
Alternative	Very favourable	Favourable	Stable	Unfavourable
Large solar farm	110	55	10	-35
Medium solar farm	82	60	28	5
Small solar farm	58	48	35	18
Partnership project	45	40	32	25

Questions/Tasks.

C4 What would the optimistic decision be (Maximax)? What would the conservative decision be (Maximin)? **[2]**

C4 - Maximax and Maximin **[2]**



Complete the table using letters first.

	A_1	A_2	A_3	A_4	Decision
Max	M_1	M_2	M_3	M_4	$A^*_{\text{Maximax}} = A_1$
Min	m_1	m_2	m_3	m_4	$A^*_{\text{Maximin}} = A_4$

Maximax

$$\begin{aligned} M_1 &= \max\{110, 55, 10, -35\} = 110 \\ M_2 &= \max\{82, 60, 28, 5\} = 82 \\ M_3 &= \max\{58, 48, 35, 18\} = 58 \\ M_4 &= \max\{45, 40, 32, 25\} = 45 \end{aligned}$$

Maximin

$$\begin{aligned} m_1 &= \min\{110, 55, 10, -35\} = -35 \\ m_2 &= \min\{82, 60, 28, 5\} = 5 \\ m_3 &= \min\{58, 48, 35, 18\} = 18 \\ m_4 &= \min\{45, 40, 32, 25\} = 25 \end{aligned}$$

Replace the letters with their values.

	A_1	A_2	A_3	A_4	Decision
Max	110	82	58	45	$A^*_{\text{Maximax}} = A_1$
Min	-35	5	18	25	$A^*_{\text{Maximin}} = A_4$

Therefore, the optimistic decision is

$$A^*_{\text{Maximax}} = A_1 \quad \text{(Large solar farm)}$$

and the conservative decision is

$$A^*_{\text{Maximin}} = A_4 \quad \text{(Partnership project).}$$